ST JOSEPH'S UNIVERSITY

BENGALURU-560027

PROJECT TITLE
Retail Sales Analysis Project

Submitted by:
Shruti Mehan, Rupali Grover
REG NO:
25PGDDA13, 25PGDDA20

Submitted to:
AARAN LAWRENCE DLIMA
Department of Advanced Computing
St. Joseph's University
36, Lalbagh Road
Bangalore- 560027

Table of content:

S no.	Topics	Page no.
1.	Introduction	1
2.	Objectives	2
3.	Tools and Technologies	2
4.	Dataset Description	3
5.	Data preprocessing: : Data type Summary : Data cleaning and Prepration	4-5
6.	Data Analysis and Visualization	5-10
7.	Key Insights	11
8.	Learnings	11
9.	Conclusion	12

1. Introduction:

In today's highly competitive and data-driven business environment, retail organizations rely heavily on data analytics to understand customer behavior, evaluate sales performance, and support strategic decision-making. The retail industry generates large volumes of data through daily sales transactions, customer purchases, and inventory movement. Proper analysis of this data helps businesses identify trends, patterns, and opportunities for growth.

Retail sales analysis plays a crucial role in helping organizations monitor performance, optimize product offerings, and improve customer satisfaction. By analyzing historical sales data, businesses can understand which products perform well, identify seasonal demand, and recognize key customer segments that contribute most to revenue. This enables management to make informed decisions related to pricing, promotions, inventory planning, and marketing strategies.

This project focuses on analyzing a retail sales dataset using Excel and Python, two widely used tools in data analytics. Excel is used for initial data cleaning, organization, and basic analysis, while Python is used for advanced data analysis and visualization. The project demonstrates a complete data analysis workflow starting from raw data preparation to generating meaningful insights through visual representations. The outcomes of this project highlight how data-driven approaches can support better business decisions and improve overall retail performance.

2. Objectives of the Project:

The main objectives of this project are:

- **To analyze overall sales performance and identify revenue trends**
- **To understand product-wise performance and identify top-selling and low-performing categories**
- **To study customer purchasing behavior based on age, age group, and gender**
- **To perform time-based sales analysis to identify monthly and yearly trends**
- **To identify patterns in customer demand and seasonal variations in sales**
- **To evaluate the relationship between quantity sold and total revenue**
- **To present insights using appropriate charts and visualizations**
- **To apply Excel and Python techniques for data cleaning, analysis, and visualization**
- **To support data-driven decision-making through meaningful insights**

3. Tools and Technologies Used:

Excel: Data cleaning, sorting, filtering, pivot tables, and charts

Python: Pandas and NumPy for data analysis, Matplotlib for visualization

4. Dataset Description:

The retail sales dataset contains transaction-level data from a retail store. The main attributes include:

- **Transaction ID – Unique identifier for each transaction**
- **Date – Date of purchase**
- **Customer ID – Unique customer identifier**
- **Gender – Gender of the customer**
- **Age – Age of the customer**
- **Age Group – Categorized age range of customers**
- **Product Category – Type of product purchased**
- **Quantity Sold – Number of units sold**
- **Unit Price – Price per unit**
- **Total Sales – Total transaction value (Quantity × Unit Price)**

5. Data Preprocessing:

1. Data Type Summary

The table below summarizes the data types of key variables used in the analysis:

- **Transaction ID: Integer / Text (Identifier)**
- **Date: Date**
- **Customer ID: Integer / Text (Identifier)**
- **Gender: Categorical (Text)**
- **Age: Numerical (Integer)**
- **Age Group: Categorical (Text)**
- **Product Category: Categorical (Text)**
- **Quantity Sold: Numerical (Integer)**
- **Unit Price: Numerical (Float)**
- **Total Sales: Numerical (Float)**

2. Data Cleaning and Preparation

Data cleaning and preparation were performed to ensure accurate and reliable analysis. The following steps were carried out:

- **Checked and corrected data types (date, numeric, and text fields)**
- **Handled missing values by removing or replacing them with appropriate values**
- **Removed duplicate transaction records to avoid incorrect sales calculations**

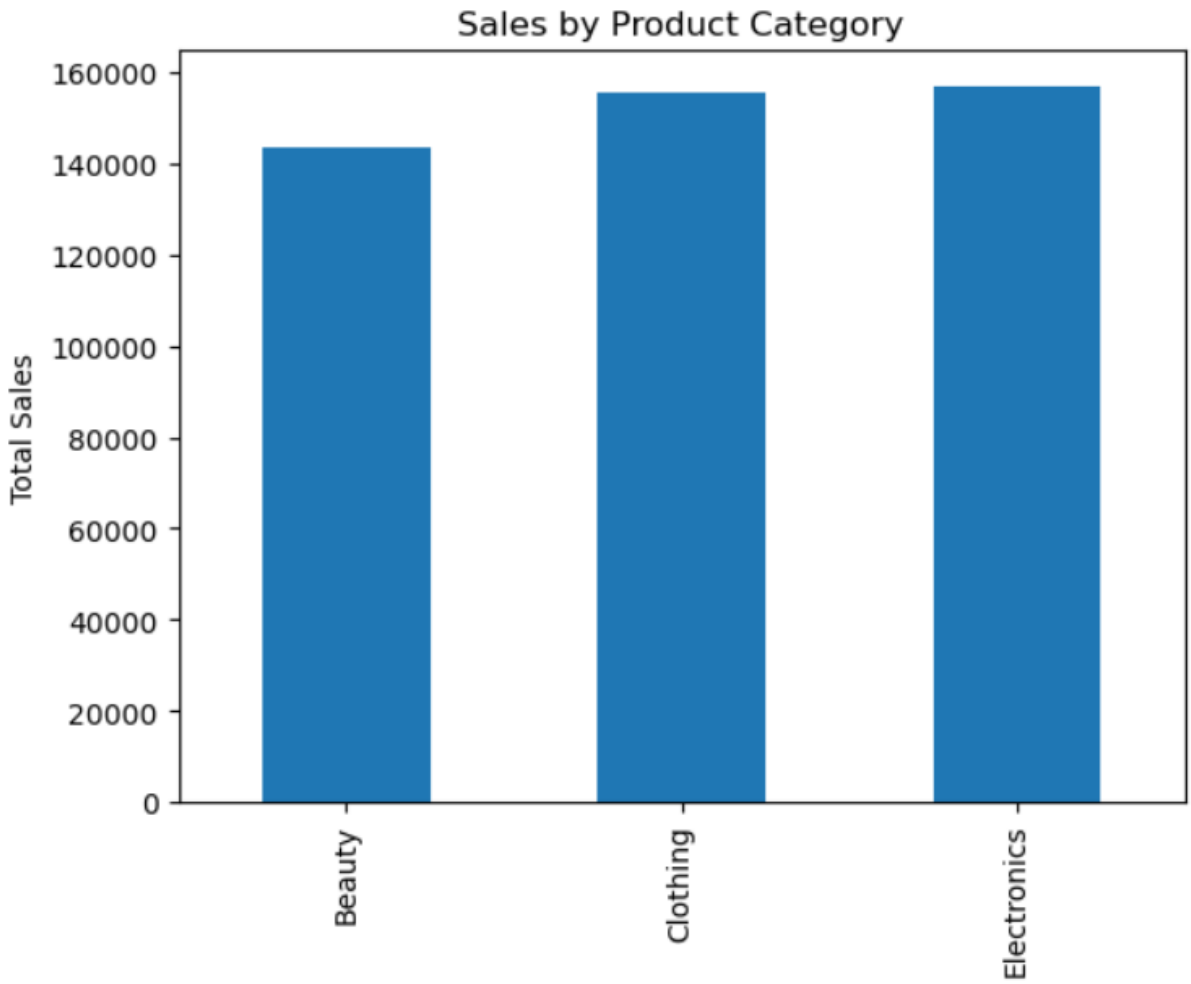
- **Created calculated fields such as Total Sales where required**
- **Ensured consistency in product category names and customer details**

Excel was used for initial data inspection and cleaning, while Python was used for further data validation and transformation.

Raw Data → Data Cleaning → Data Transformation → Feature Creation → Final Dataset

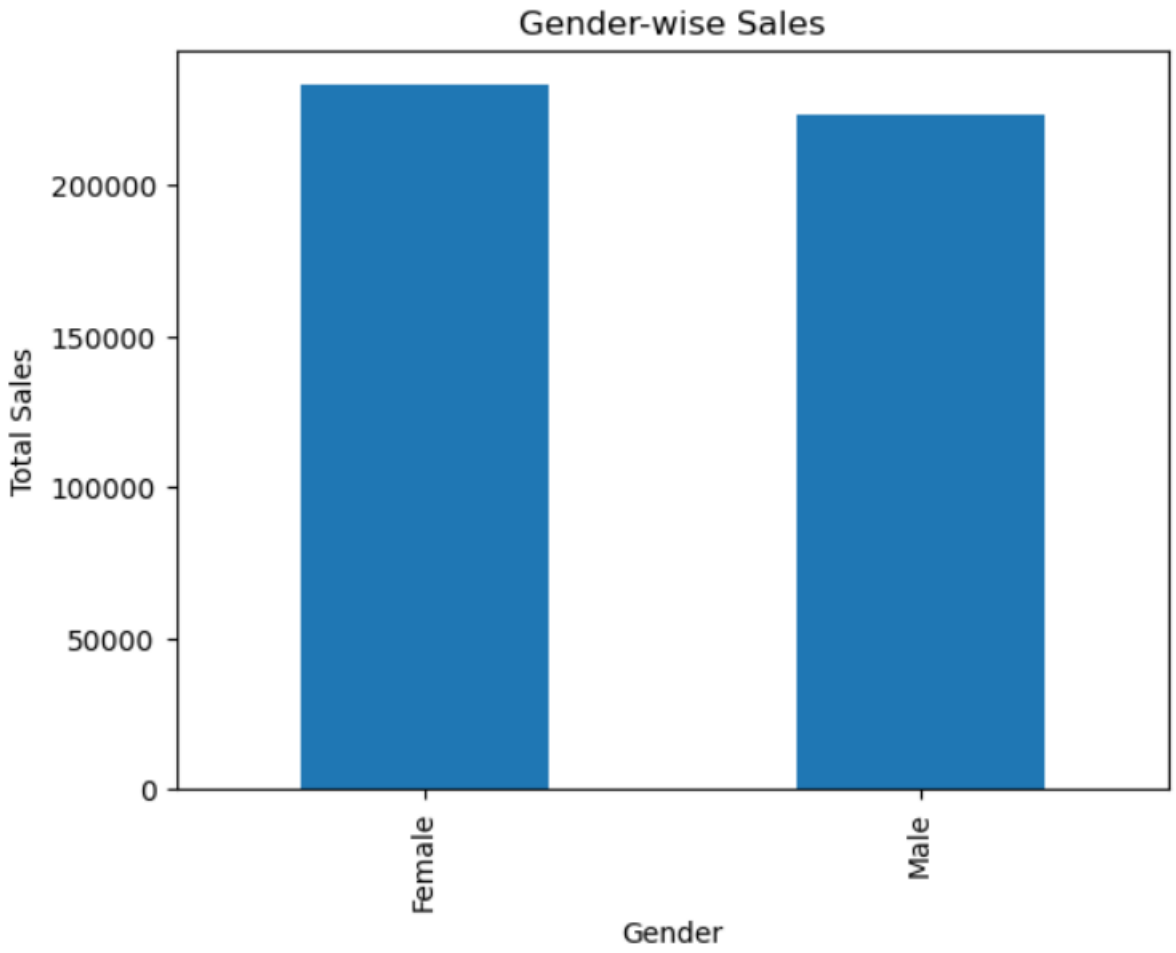
6. Data Analysis and Visualization:

Title: Sales Performance by Product Category



Explanation:
This bar chart illustrates the total sales generated by different product categories, namely Beauty, Clothing, and Electronics. From the visualization, it is evident that Electronics recorded the highest total sales, followed closely by Clothing, while Beauty products contributed comparatively lower sales. This comparison helps identify high-performing product categories and provides valuable insights for inventory planning, promotional strategies, and resource allocation. Retail managers can use this information to focus on categories that drive maximum revenue while developing strategies to improve the performance of lower-selling categories.

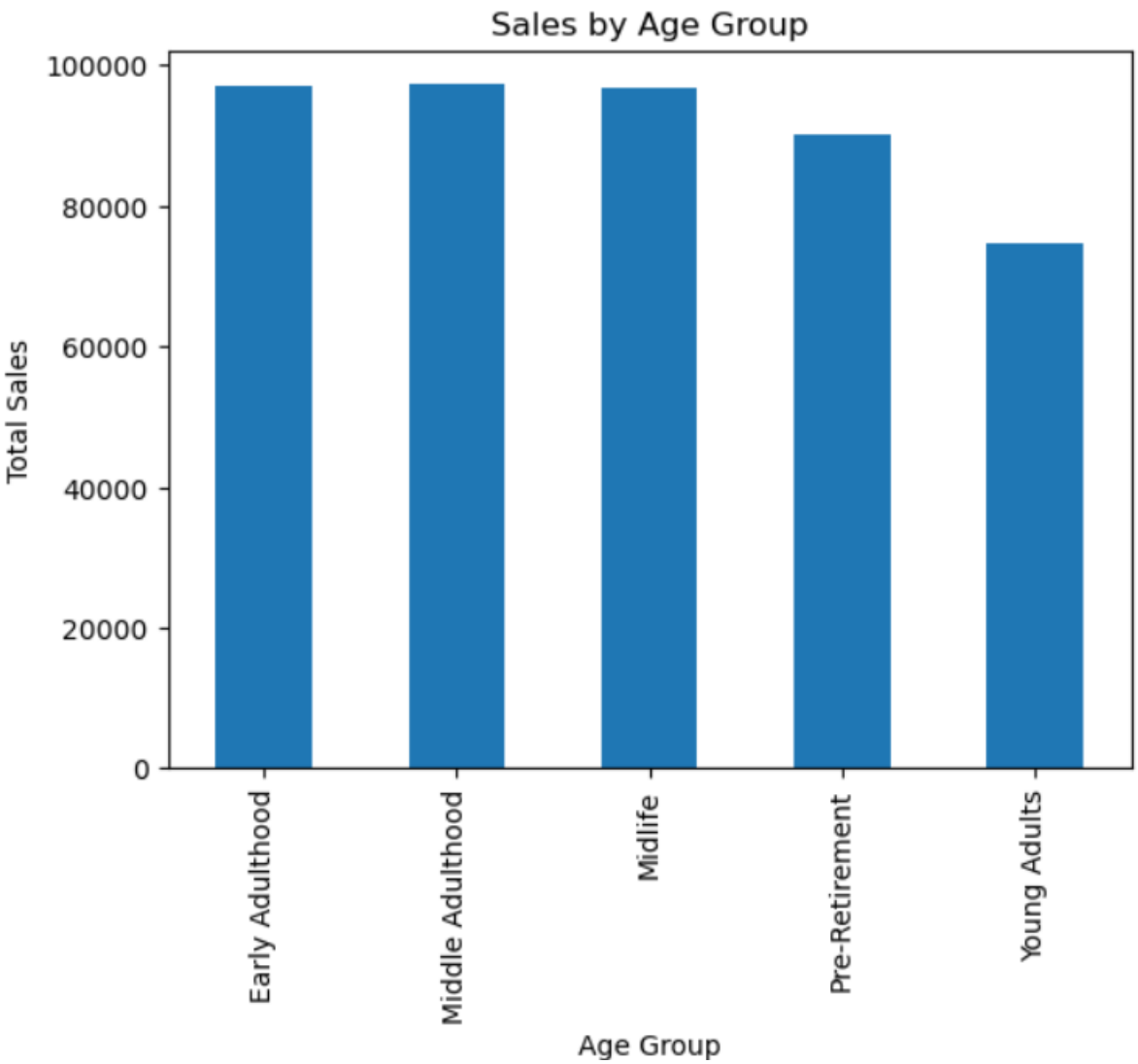
Title: Gender-wise Sales Distribution



Explanation:

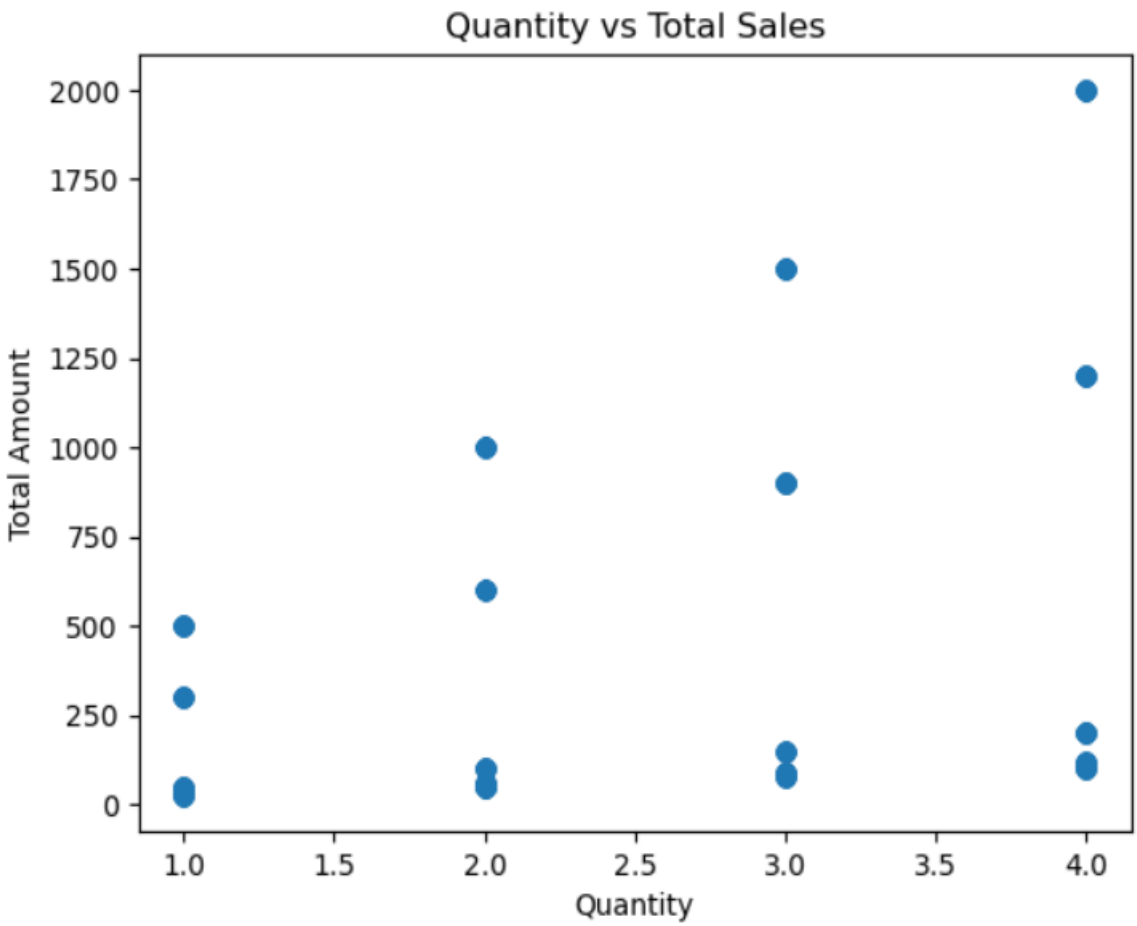
This bar chart represents the total sales generated by customers based on gender. The visualization shows that female customers contributed slightly higher total sales compared to male customers. This indicates that female shoppers play a significant role in overall revenue generation. Such insights can help retailers design gender-specific marketing strategies, personalized promotions, and product assortments to further enhance sales performance.

Title: Sales Distribution Across Different Age Groups



This bar chart shows the total sales generated by customers across various age groups. The visualization indicates that customers in the Early Adulthood, Middle Adulthood, and Midlife categories contribute the highest sales, highlighting these groups as the primary revenue drivers. In contrast, Young Adults and Pre-Retirement age groups contribute relatively lower sales. This analysis helps retailers understand purchasing power across age segments and enables targeted marketing strategies tailored to high-value age groups.

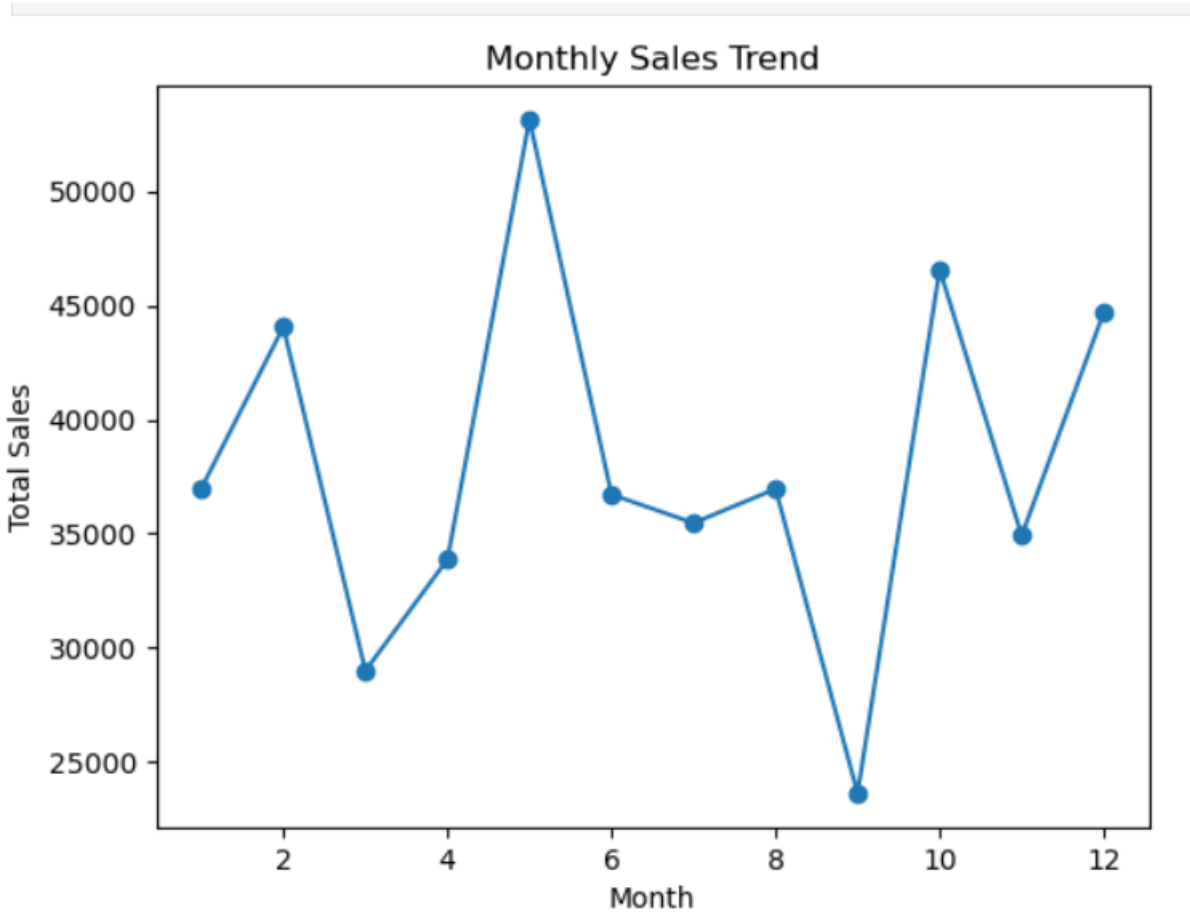
Title: Relationship Between Quantity Sold and Total Sales



Explanation:

This scatter plot illustrates the relationship between the quantity of products sold and the corresponding total sales amount. The visualization shows a positive relationship between quantity and total sales, indicating that as the number of units sold increases, the total sales value generally increases. However, variations in total sales for the same quantity level suggest differences in unit prices across products. This analysis helps in understanding how both quantity and pricing influence overall revenue.

Title: Monthly Sales Trend Analysis



Explanation:
This line chart represents the monthly trend of total sales over the year. The visualization shows noticeable fluctuations in sales across different months, indicating the presence of seasonal patterns in customer purchasing behavior. Sales peak around certain months, while a significant drop is observed in some periods, suggesting lower demand during those months. This trend analysis helps retailers understand seasonality, plan inventory levels, schedule promotions, and allocate resources more effectively throughout the year.

7. Key Insights:

Certain product categories contributed significantly more to total revenue than others

Sales showed noticeable variation across different months, indicating seasonal trends

Customers from specific age groups generated higher sales compared to others

A relationship was observed between quantity sold and total revenue

A small group of customers contributed a large portion of total sales

8. Learnings:

- **Gained experience in completing an end-to-end data analysis project**
- **Improved ability to present insights clearly using charts and structured reporting**
- **Developed confidence in applying Excel and Python together for real-world problems**

9. Conclusion:

This Retail Sales Analysis project demonstrated how Excel and Python can be effectively used to clean data, analyze trends, and visualize business insights. The analysis provided valuable information about product performance, customer behavior, and sales trends that can help management make informed decisions and improve sales strategies.